On Convergences beyond Zero-Sum Markov Games

A self-contained reading guide for Markov games and online learning

Po-An Chen Chi-Jen Lu Chuang-Chieh Lin Wei-Chen Lin
Chih-Chieh Hung

NYCU Academia Sinica NTOU NCHU

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Outline

What is this project about?

One-line summary

We study what can still be guaranteed when several strategic learners interact repeatedly in a stochastic environment, beyond the well-understood zero-sum case.

Markov game → Online learning → Equilibrium / PoA

The paper asks: if agents use no-regret or optimistic gradient-type updates, do their policies become meaningful in a general-sum Markov game?

Why "beyond zero-sum" matters

Zero-sum Markov games

- Player 1's gain is player 2's loss.
- Minimax structure gives strong stability.
- Many algorithms have convergence guarantees.

General-sum Markov games

- Agents may partially cooperate or conflict.
- Nash equilibria may be hard to reach.
- Learning dynamics may cycle or behave irregularly.

Main difficulty

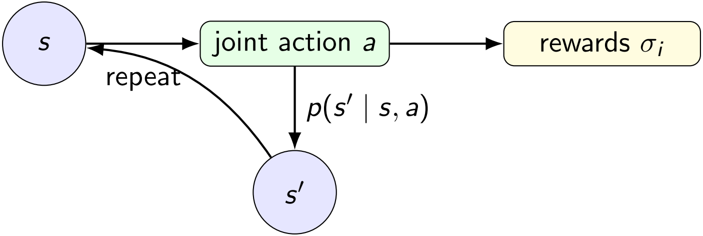
There is usually no single global objective and no minimax saddle-point structure.

Two guarantees pursued in the work

1. **Generic no-regret learning in smooth Markov games.**
   - The time-average welfare is close to the optimal welfare up to a robust price-of-anarchy factor.
2. **Optimistic Gradient Ascent (OGA) in general-sum Markov games.**
   - Either some iterate is an approximate Nash equilibrium, or the average welfare is still good.

No-regret learners → Smoothness → Bounded PoA

A useful mental model



A Markov game is a repeated strategic interaction where the current state affects the available rewards and the next state.

Ingredients of a Markov game

A finite discounted Markov game consists of:

- a finite state set $\mathcal{S}$;
- players $i \in \mathcal{N}$;
- action sets $\mathcal{A}_i$ and joint action $a = (a_i)_i$;
- transition probability $p(s' \mid s, a)$;
- reward function $\sigma_i(s, a)$ for each player;
- discount factor $\gamma \in [0, 1)$.

Interpretation

At each state s, all players choose actions simultaneously. Then everyone receives rewards and the system moves to a new state.

Policies: strategies that depend on the state

Definition (Stationary policy)

A stationary policy of player i maps every state s to a distribution over actions:

$$\pi_i(s) \in \Delta(\mathcal{A}_i).$$

A joint policy is $\pi = (\pi_1, \ldots, \pi_n)$.

state s $\longrightarrow$ policy $\pi_i(s)$ $\longrightarrow$ action distribution

Value: the long-run objective of each player

For a joint policy π and initial state s, player i's discounted value is

$$V_i^{\pi}(s) = \mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^{t-1} \sigma_i(s_t, a_t) \mid s_1 = s,\ a_t \sim \pi(s_t)\right].$$

Plain-language meaning

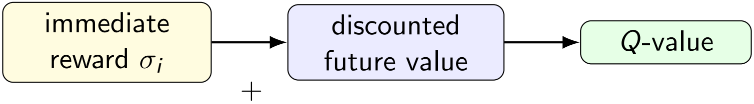
The value is the expected total reward, where future rewards are discounted by γ.

- Larger γ: future states matter more.
- Smaller γ: near-term rewards dominate.

Q-values: one-step deviation plus future value

For two players with mixed actions x^s, y^s at state s, the manuscript defines a state-action value matrix:

$$Q^s_{i,t}(a,b) := \sigma_i(s,a,b) + \gamma\,\mathbb{E}_{s' \sim p(\cdot|s,a,b)}\big[V^{s'}_{i,t-1}\big].$$



A Q-matrix turns the dynamic Markov game locally into a matrix game at state s.

Nash equilibrium policy

Definition (Nash policy)

A joint policy π is a Nash equilibrium if no player can improve its value by unilaterally changing its own policy:

$$V_i^{\pi}(s) \geq V_i^{\pi_i', \pi_{-i}}(s) \quad \text{for all } i, \pi_i', s.$$

Important point

The deviation is a full policy deviation, not merely a one-shot action deviation.

Approximate Nash equilibrium

Definition (ε-approximate Nash)

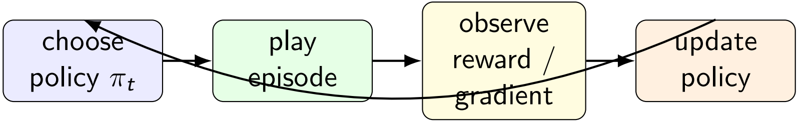
A joint policy π is an ε-approximate Nash equilibrium if every player can gain at most ε by deviating:

$$V_i^{\pi'_i,\pi_{-i}}(s) - V_i^{\pi}(s) \leq \varepsilon \quad \textit{for all } i, \pi'_i, s.$$

- Exact equilibrium may be too hard to reach.
- Approximate equilibrium is the natural algorithmic target.
- Smaller ε means stronger convergence.

Repeated learning view

The manuscript studies repeated play over iterations or episodes.



Goal

Analyze what the sequence $\pi_1, \pi_2, \ldots, \pi_T$ implies about equilibrium quality or social welfare.

Regret: comparison with the best fixed deviation

For episodic Markov games, the regret of player i is

$$R_i^T = \max_{\pi_i} \sum_{t=1}^{T} V_i^{\pi_i, \pi_{-i}^t}(s) - \sum_{t=1}^{T} V_i^{\pi^t}(s).$$

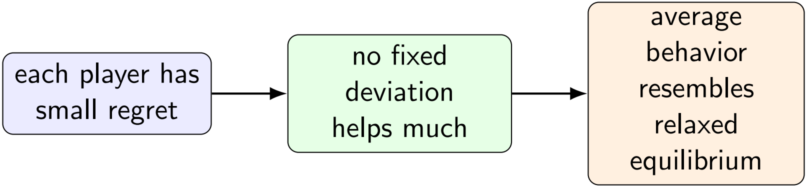
No-regret property

Player i is no-regret if

$$R_i^T = o(T).$$

Equivalently, average regret $R_i^T / T \to 0$.

Why no-regret learning is useful



In normal-form games, no-regret learning implies convergence of empirical distributions to coarse correlated equilibrium. The project asks what analogous welfare and equilibrium statements can be made for Markov games.

Last iterate vs. time average

Time-average guarantee

$$\frac{1}{T}\sum_{t=1}^{T}\pi_t$$

The average behavior is good.

Last-iterate guarantee

$$\pi_T$$

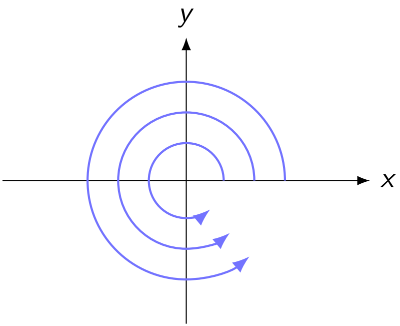
The final produced policy is good.

Hierarchy

Last-iterate convergence is usually much stronger and harder than average convergence.

Why ordinary gradient dynamics may fail

In games, each player updates against other players who are also moving.



rotational behavior / cycling

Even bilinear zero-sum games may exhibit cycling under basic gradient descent-ascent. Optimism is one standard remedy.

Social welfare and optimal policy

Define social welfare at state s by

$$V_\pi(s) = \sum_i V_i^\pi(s).$$

Let π^* be an optimal joint policy:

$$\pi^* \in \arg\max_\pi V_\pi(s).$$

Question

If players learn selfishly, how much social welfare can we still guarantee compared with $V_{\pi^*}(s)$?

Price of Anarchy: the welfare-loss perspective

Definition (Informal price of anarchy)

$$\mathrm{PoA} \approx \frac{\textit{welfare of an optimal solution}}{\textit{welfare obtained under strategic behavior}}.$$

central planner chooses π^* $\longleftrightarrow$ welfare gap $\longleftrightarrow$ selfish learners choose π_t

A bounded PoA says selfish learning does not destroy too much welfare.

Smoothness: the key structural assumption

Definition ((α, β)-smooth Markov game)

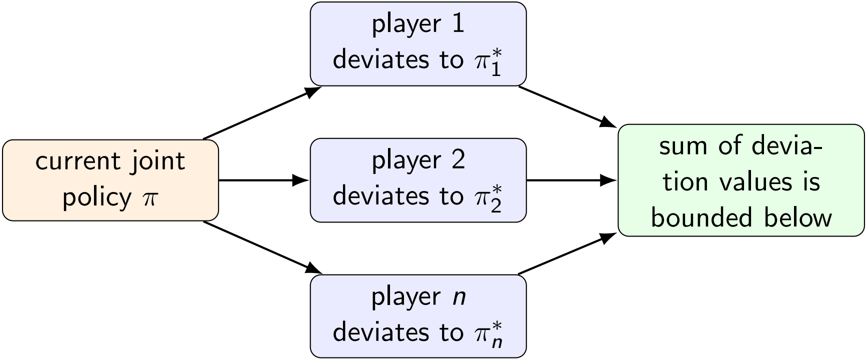
For every pair of policies π, π^* and every state s,

$$\sum_i V_i^{\pi_i^*, \pi_{-i}}(s) \geq \alpha V_{\pi^*}(s) - \beta V_{\pi}(s).$$

Interpretation

Even if every player only switches to its own component of the optimal policy while others remain at π_{-i}, the sum of these deviation values remains large.

Smoothness as a deviation argument



This is the mechanism that converts individual no-regret guarantees into a welfare guarantee.

Theorem 1: generic no-regret learners

Theorem (No-regret learning in smooth Markov games)

If the Markov game is (α, β)-smooth and every player has no regret, then for any initial state s,

$$\sum_i \frac{1}{T} \sum_{t=1}^{T} V_i^{\pi_t}(s) + \sum_i o(1) \geq \frac{\alpha}{1+\beta} V_{\pi^*}(s).$$

Message

The average welfare of selfish no-regret learning is at least an $\alpha/(1+\beta)$ fraction of optimal welfare, asymptotically.

Proof idea of Theorem 1

1. No-regret: for each player, actual cumulative value is almost as good as deviating to π_i^*.

$$\sum_t V_i^{\pi_t}(s) + o(T) \geq \sum_t V_i^{\pi_i^*, \pi_{-i}^t}(s).$$

2. Sum over all players.
3. Apply (α, β)-smoothness at each episode t.
4. Rearrange the term involving $\beta \cdot V_{\pi_t}(s)$.

Important

The theorem does not require a particular no-regret algorithm.

Why study a specific algorithm?

Theorem 1 is broad but gives only an average welfare guarantee.

Question

Can a specific learning dynamics produce an approximate equilibrium iterate in a general-sum Markov game?

Candidate dynamics

Optimistic Gradient Ascent (OGA), inspired by OGDA / optimistic mirror descent methods used in saddle-point and game learning.

OGA intuition: update using a prediction

Ordinary gradient update:

$$x_{t+1} = \Pi\{x_t + \eta g_t\}.$$

Optimistic update uses an extra look-ahead style step:

$$\widehat{x}_{t+1} = \Pi\{\widehat{x}_t + \eta g_t\}, \qquad x_{t+1} = \Pi\{\widehat{x}_{t+1} + \eta g_t\}.$$

$$\widehat{x}_t \xrightarrow{+\eta g_t} \widehat{x}_{t+1} \xrightarrow{+\eta g_t} x_{t+1}$$

OGA in the two-player Markov game

At each state s, the updates are

$$\widehat{x}_{t+1}^s = \Pi_{\Delta_X}\{\widehat{x}_t^s + \eta u_t^s\}, \quad x_{t+1}^s = \Pi_{\Delta_X}\{\widehat{x}_{t+1}^s + \eta u_t^s\},$$

$$\widehat{y}_{t+1}^s = \Pi_{\Delta_Y}\{\widehat{y}_t^s + \eta r_t^s\}, \quad y_{t+1}^s = \Pi_{\Delta_Y}\{\widehat{y}_{t+1}^s + \eta r_t^s\}.$$

Here u_t^s approximates $Q_{1,t}^s y_t^s$, and r_t^s approximates $(x_t^s)^\top Q_{2,t}^s$.

What are the gradient estimators?

For player 1, fixing player 2's mixed action y_t^s,

$$Q_{1,t}^s y_t^s$$

is the vector of expected payoffs for player 1's possible actions.
For player 2, fixing player 1's mixed action x_t^s,

$$(x_t^s)^\top Q_{2,t}^s$$

is the vector of expected payoffs for player 2's possible actions.

In the manuscript

The estimators satisfy approximation bounds such as $\|u_t^s - Q_{1,t}^s y_t^s\| \leq \varepsilon$.

The technical role of Q-value stability

The OGA analysis needs to control changes in the local matrix games:

$$\|Q_{i,t}^s - Q_{i,t-1}^s\|.$$

Why this is hard

Q-values depend on future values, which depend on policies at all states. A small local policy change can propagate through the Markov chain.

policy change → value change → Q-matrix change

Theorem 2: a dichotomy

Theorem (Informal version)

Suppose players use OGA with suitable norms, smooth regularizers, and learning rate $\eta \leq O(1/\sqrt{T})$*. Then, after sufficiently many iterations, one of two things happens:*

1. *some iterate is an approximate Nash equilibrium; or*
2. *the average welfare satisfies a bounded price-of-anarchy style guarantee.*

Message

If OGA does not expose a nearly stable point, the cumulative movement is large enough to certify a welfare lower bound.

Theorem 2: more mathematical statement

For any $\varepsilon > 0$, after

$$T > \frac{2}{\varepsilon^2 |\mathcal{S}|} \sum_{s \in \mathcal{S}} \sum_{i} \Omega_i^s$$

iterations, there exists an iterate z_t^s that is an approximate Nash equilibrium with error on the order of

$$\varepsilon \left(C^* \max\{\| Q_{1,t}^s y_t^s \|_\infty, \|(x_t^s)^\top Q_{2,t}^s \|_\infty\} + \frac{2 \max\{G_1 \Omega_1^s, G_2 \Omega_2^s\}}{\eta} \right),$$

or else the average welfare has a PoA-type lower bound.

Core proof object: total regret plus movement

The proof bounds the sum of regrets by

$$R_1^T + R_2^T \le \frac{1}{2\eta}\|\widehat{z}_1 - z^*\|^2 - \frac{1}{4\eta|\mathcal{S}|}\sum_{s\in\mathcal{S}}\sum_{t=1}^{T}\left(\|\widehat{z}_{t+1}^s - z_t^s\|^2 + \|z_t^s - \widehat{z}_t^s\|^2\right).$$

Interpretation

If the movement terms are large for all t, the total regret becomes negative enough to imply strong welfare. If some movement term is small, that point is close to stationary and yields approximate Nash.

The small-movement branch

Suppose there exists t such that

$$\|\widehat{z}^s_{t+1} - z^s_t\|^2 + \|z^s_t - \widehat{z}^s_t\|^2 \leq \varepsilon^2.$$

Then the two OGA points are close.

$$\widehat{z}_t \xleftrightarrow{\text{small}} z_t \xleftrightarrow{\text{small}} \widehat{z}_{t+1}$$

Small movement plus the variational inequality form of the update implies no player has a large profitable deviation.

The large-movement branch

Suppose instead every iteration has movement larger than ε^2. Then the negative movement term accumulates:

$$R_1^T + R_2^T \leq -\frac{\varepsilon^2 T}{8\eta}.$$

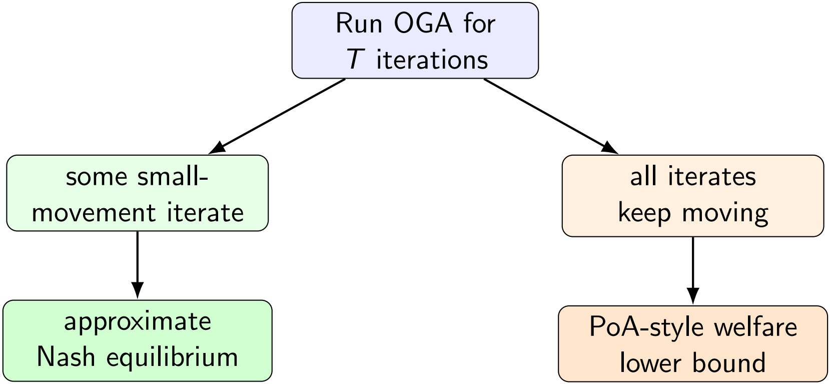
Combining this with the smoothness/PoA argument yields

$$\frac{1}{T}\sum_{t=1}^{T} V_{\pi_t}(s) \geq \frac{\alpha}{1+\beta} V_{\pi^*}(s) + \frac{1}{1+\beta}\frac{\varepsilon^2}{8\eta}.$$

Intuition

Persistent movement prevents equilibrium convergence but implies the sequence extracts high welfare under smoothness.

Dichotomy in one diagram



What should a new coauthor understand first?

1. Markov games: states, policies, values, Q-values.
2. No-regret learning: why individual regret implies collective guarantees.
3. Smoothness: how deviation inequalities imply robust PoA.
4. OGA: why the two-step update helps control cycling.
5. Dichotomy proof: small movement gives approximate Nash; large movement gives welfare.

Potential weak points to check carefully

Mathematical details worth auditing

- Exact definition of smoothness for discounted versus episodic Markov games.
- Whether the stated regret notion matches policy deviations over full trajectories.
- Whether $\sum_i R_i^T \geq 0$ or other sign assumptions are justified.
- Dependence on γ, $|\mathcal{S}|$, estimator error, and norm constants.
- Extension from two players to multiple players for OGA.

Possible extension directions

1. **Equilibrium concepts:** extend Theorem 1 to mixed, correlated, and coarse correlated policy profiles.
2. **Sharper convergence:** improve the approximate Nash error or prove last-iterate convergence under stronger assumptions.
3. **Game classes:** identify concrete Markov games satisfying (α, β)-smoothness.
4. **Algorithms:** compare OGA, optimistic mirror descent, extra-gradient, and multiplicative weights.
5. **Experiments:** test whether the dichotomy is visible in small general-sum Markov games.

Takeaway

Main conceptual contribution

The work gives two ways to extract meaningful guarantees from learning in general-sum Markov games: average welfare via smoothness and a dichotomy guarantee via OGA.

Main technical challenge

General-sum Markov games lack the minimax structure of zero-sum games, so convergence must be obtained through weaker notions, stronger assumptions, or carefully designed dynamics.

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